

Finfluencing Portfolio Asset Allocation

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Abstract. In the United States, social media and large language models (LLMs) have become a key source of information for financial decision-making. Reddit, an online forum centered on communities called subreddits, attracted public attention during the meme-stock craze of 2021, becoming a new mainstream outlet for financial information. In a crowd-sourced experiment to observe whether Reddit posts and LLM recommendations can influence the asset class allocations of a hypothetical investment portfolio, this study typically found a statistically significant difference in the allocation weight of the subject asset class before and after showing a participant social media or LLM generated content related to it. The study found that exposure to top posts from the r/StockMarket Reddit community led to a shift in portfolio allocation from stocks to cash, whereas exposure to top posts in the r/CryptoMarkets community led to increased allocation to crypto and reduced exposure to bonds. Showing subjects a prompt and a response from various LLMs asking which asset class is best to invest in with all responses advising stocks, led participants to increase their exposure to stocks while reducing their allocations to cash and crypto, contrary to our findings using recent social media posts.

Keywords: portfolio allocation · social media · large language models

1 Introduction

Asset allocation is a tool used in portfolio management to manage returns versus risk. Financial institutions offer retail investors many strategies and products centered on this concept, such as the 60/40 balanced portfolio, target-date funds, and robo-advisors, to help individuals reach their investment goals. Given the wealth of options, individuals also seek advice from a variety of sources, including professional financial advisors and family members. However, with the prevalence of social media and artificial intelligence in American society, individual investors are now increasingly seeking financial advice from these sources. In 2021, TIAA [13] found that 11% of 1,000 online survey respondents most often rely on social media for information about financial decisions, and that 32% would trust the financial advice of social media influencers and celebrities. According to a more recent Bankrate [2] analysis using data taken in 2023,

approximately 30% of Americans use social media for financial advice, while American Banker [1] reports that approximately 55% have used AI for the same purpose. Another study by Pak [6] surveying adults in South Korea found that 67.8% of 2,170 respondents have used LLMs for at least one financial task, with the most common being stock investment (50.3%), affirming the role LLMs play in providing investment advice.

By using social media, large language models (LLMs), and crowdsourcing, this paper seeks to answer the following questions:

- RQ1: Do posts from social media influence people in changing their portfolio asset allocation preferences?
- RQ2: Can information produced from LLM’s influence people in changing their portfolio asset allocation preferences?
- RQ3: If there is a reallocation to one asset due to social media or LLM influence, how are the other assets re-weighted to accommodate the change?

Answering these questions can help deepen our understanding of behavioral finance in the age of social media and artificial intelligence. Investment firms and financial advisors can gain insight into how social media and LLMs can be used to introduce new investment products and services to the public. In addition, it can help individual investors understand how they are influenced into making investment decisions.

2 Background

The study of social media’s effects on investment decision-making has been subject to a comprehensive literature review by Singh and Chakraborty [12], who classified 112 journal articles published between 2013 and 2023 along the dimensions of theories, context, characteristics, and methodologies. The review highlights the interest and diversity of studies conducted in this field.

Related work on the use of social media to influence investment decisions includes a study by Kuerzinger and Stangor [4] that examines the impact of AI-generated 140-character tweets presented alongside financial information on investment amounts. They found evidence of a causal effect of social media sentiment on their subjects’ behavior. Another relevant study by Sathya and Prabhavathi [10] surveyed participants on their social media usage and their investment decision-making, such as investment goals, investment horizon, and preferred investment products, to find a correlation between social media usage and investment decisions.

Other related work in the field includes a method for using Twitter data to create a sentiment score that can help guide investment decisions Hasselgren et al. [3]. While another study by Wang and Gu [17] tackles investment decision bias from two fronts. They discovered that fund selections remained stable across demographic groups, but investment amounts differed when investor names signaled race or gender, controlling for age and income. Wang and Gu also found that allocations varied by the fund manager’s gender and race.

Unlike Kuerzinger and Stangor’s study and Sathya and Prabhavathi’s work, this report plans to use authentic social media posts with high engagement from

Reddit. This study differs in that it attempts to measure differences in portfolio allocation before and after a user is shown LLM-generated recommendations for a particular asset.

3 Methodology

3.1 Data Collection

Social Media Posts Social media posts are collected from Reddit across the top investing-related subreddits focused on a single asset class: stocks, bonds, and cryptocurrency. In this study, we source posts from r/StockMarket, r/bonds, and r/CryptoMarkets because they are the largest subreddits with a broad market focus within their respective asset classes, with a 28-day rolling average weekly visitor volume of 1M, 26K, and 116K at the end of April 2026, respectively [9] [8] [7]. For each of these subreddits, we look back up to one year from April 2026 and collect the top one-hundred posts, defined as those with the highest net score of upvotes and downvotes. Within these posts, we collect the top five that do not contain videos or a removed cross-post, and then screenshot each post along with its most upvoted comment.

LLM Recommendations To simulate what a potential investor might ask an LLM in terms of investment advice for portfolio allocation, the study provided the following prompt to five different LLMs: "What is the best asset class to invest in out of stocks, bonds, crypto, and cash? Give a one-paragraph recommendation". Screenshots of the results were collected from five LLMs: Anthropic Claude Sonnet 4.6, OpenAI GPT-5.3, Google Gemini 3, Microsoft Copilot, and xAI Grok.

Portfolio Allocation Survey A Qualtrics survey was constructed using the top social media posts for each asset class and the LLM asset class recommendation. The survey participant is presented with a scenario in which they allocate a percentage of \$10,000 in a portfolio among four asset classes: stocks, bonds, alternatives (e.g., crypto), and cash. The survey participant is shown screenshots of either five social media posts from a single asset-focused subreddit or five LLM-generated asset class recommendations, and then prompted to allocate their portfolio again.

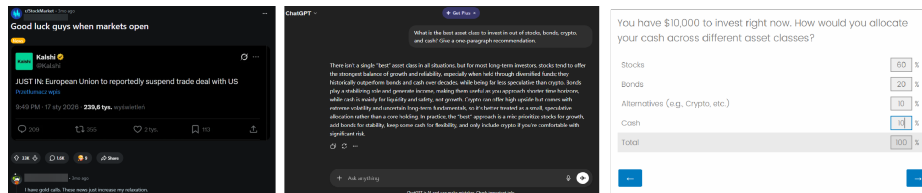


Fig. 1: From left to right, example posts shown to participants with usernames redacted [15] [16], example LLM (ChatGPT) prompt and response displayed to participants [5], and portfolio allocation question presented to participants in Qualtrics

Crowdsourcing Pre/Post Portfolio Allocation Survey Data One hundred respondents, twenty-five for each asset class and the LLM experiment, were crowdsourced from two platforms: Amazon Mechanical Turk and Prolific. Amazon Mechanical Turk (<https://www.mturk.com/>) is a platform where requestors

can outsource tasks, called Human Intelligence Tasks (HITs), to workers for a fee. In this study, Mechanical Turk participants were filtered according to criteria such as being located in the United States, a Mechanical Turk Master’s qualification, an approval rating of $\geq 95\%$, and more than 10,000 completed HITs to ensure high-quality respondents.

Prolific (<https://www.prolific.com/>) is another crowdsourcing platform that connects researchers with high-quality participants by imposing stricter account verification, implementing bot detection, and enforcing a minimum ethical wage. Participants recruited from Prolific for this study are also required to meet the criterion of residing in the United States.

Within the Qualtrics survey, reCAPTCHA bot detection and questions about the social media or LLM content help further mitigate bots and encourage participant engagement. This study did not post surveys simultaneously across platforms, and participants on one platform were limited to participating in only one survey. Approximately half of the study’s sample is sourced from Mechanical Turk and half from Prolific, with the assumption that a Master Qualified worker from the former platform is equivalent in quality to one from the latter for the purposes of this study.

3.2 Data Analysis

Reddit Examining Table 1, we observe a large difference in weekly visitors and contributions to the asset class-focused subreddits. The community r/StockMarket has more than eight times the visitors (1M) than the next closest subreddit, r/CryptoMarkets (116K), but only about 25% more contributions. This indicates less active engagement among visitors to the r/StockMarket subreddit than those visiting r/CryptoMarkets. There is an overrepresented interest in crypto markets on Reddit relative to the size of the US stock market, with the US stock market having a market capitalization of \$62.2T [11] and the crypto markets having \$2.4T [14] in 2024. The r/bonds community is by far the smallest, although the market capitalization of US bonds was \$58.2T in 2024 [11], nearly the size of the US stock market.

When we examine the mean values for the top five posts by post score for each subreddit in the last year, we also observe that r/StockMarket has significantly higher average comment counts (2,837.20) and post score (40,617.60) than r/bonds and r/CryptoMarkets, each with 193.4 and 490.20 average comments along with 1,478.20 and 1,328.00 average post scores, respectively. Despite r/CryptoMarkets possessing higher weekly visitors, contributions, and average number of comments for the top five posts, the average post score of the top five comments for r/bonds is slightly higher than the average of r/CryptoMarkets, hinting at higher user engagement in terms of upvotes but lower high-level user engagement in terms of comments.

Table 1: Subreddit Statistics [9] [8] [7]

Subreddit	Weekly (28-day MA)		Top 5 Posts Past Year (Mean)		
	Visitors	Contributions	# of Comments	Post Score	Upvote Ratio
r/StockMarket	1M	4.4K	2,837.20	40,617.60	0.93
r/bonds	20K	279	193.4	1,478.20	0.95
r/CryptoMarkets	116K	3.5K	490.20	1,328.00	0.91

Crowdsourced Participants Regarding survey participants, there were no demographic constraints beyond residency in the United States on both the Amazon Mechanical Turk and Prolific platforms, given the limits of the study’s resources. As shown in Table 2, each survey consisted of twenty-five participants, and the average participant was middle-aged, either between 35-44 or 45-54 years old. Given that both platforms require a certain level of computer savvy, participants, on average, had a Bachelor’s degree, except for those participating in the r/StockMarket survey, whose average education level was an Associates or technical degree.

Table 2: Participant Statistics

Survey Study	# of Participants	Age Range (Mean)	Education Level (Mean)
r/StockMarket	25	35-44 years old	Associates or technical degree
r/bonds	25	45-54 years old	Bachelor’s degree
r/CryptoMarkets	25	45-54 years old	Bachelor’s degree
LLMs	25	35-44 years old	Bachelor’s degree
All Studies	100	35-44 years old	Bachelor’s degree

The small sample size of the study is a limitation that can increase the effects of non-representative data. Another limitation noted earlier is that r/StockMarket participants have a lower average education level than other study participants. Educational differences may lead to varying levels of exposure to financial and investment concepts. This may affect participants’ understanding of the financial information described in the social media posts and their portfolio allocation preferences across asset classes.

4 Findings and Results

4.1 Stocks - Portfolio Allocations Pre/Post Reddit Influence

In performing a two-sided paired t-test at a 95% confidence level for each asset class allocation in the portfolio before and after showing content from the r/StockMarket subreddit, we found that the asset allocation difference is statistically significant for both stocks and cash as depicted in Table 3. Based on Figures 2 and 3, we observe that on average, participants reduced their stock allocation and increased their cash allocation. We infer that the Reddit posts had a negative impact on stock allocation, driven by negative sentiment regarding the stock market. The reallocation of stocks into cash suggests a flight to safety.

Table 3: Portfolio Allocation Statistics: Pre/Post reading posts from r/StockMarket

Asset Class	Pre-Reddit allocation		Post-Reddit allocation		t-statistic	p-value
	Mean	Median	Mean	Median		
Stocks	0.4512	0.5000	0.3512	0.3000	2.9382	0.0072
Bonds	0.1760	0.2000	0.1560	0.2000	1.1711	0.2531
Crypto	0.1360	0.0100	0.1480	0.1000	-0.6999	0.4907
Cash	0.2368	0.2000	0.3448	0.2000	-2.3630	0.0266

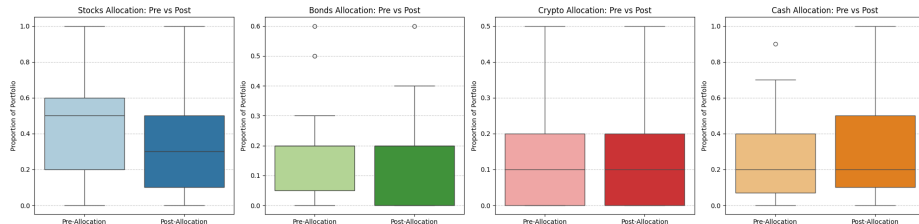


Fig. 2: Pre/Post reading Reddit Posts from r/StockMarket Portfolio Allocations

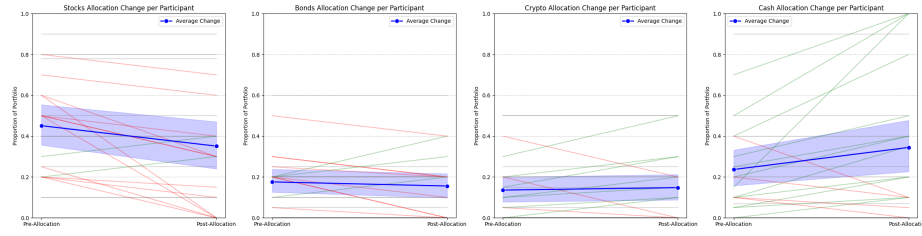


Fig. 3: Pre/Post reading Reddit Posts from r/StockMarket Portfolio Allocations

4.2 Bonds - Portfolio Allocations Pre/Post Reddit Influence

When analyzing the r/bonds subreddit results of the two-sided paired t-test at a 95% confidence level for each asset class allocation in Table 4, we observe only a statistically significant difference in the allocation of stocks with a small decline in portfolio allocation. However, examining the boxplot of bond allocations in Figure 4, we see that the upper and lower quartiles of bonds expanded almost symmetrically after participants were shown social media content from r/bonds. These observed changes in portfolio allocation can indicate conflicting sentiment regarding bonds where different participants might put more weight on a particular negative or positive post than another.

Table 4: Portfolio Allocation Statistics: Pre/Post reading posts from r/bonds

Asset Class	Pre-Reddit allocation		Post-Reddit allocation		t-statistic	p-value
	Mean	Median	Mean	Median		
Stocks	0.4284	0.3500	0.3704	0.3000	2.7211	0.0119
Bonds	0.2328	0.2000	0.232	0.2000	0.0277	0.9781
Crypto	0.0508	0.0100	0.0544	0.0500	-0.3993	0.6932
Cash	0.2880	0.2000	0.3432	0.2500	-2.0005	0.0569

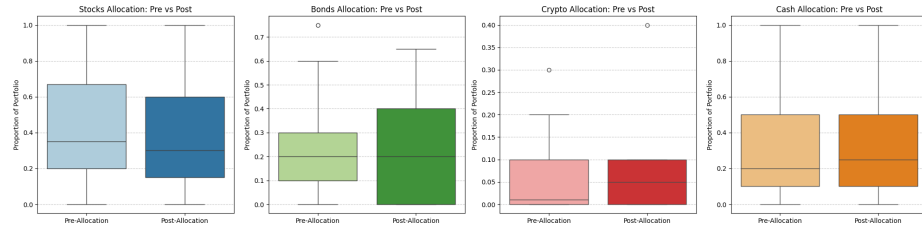


Fig. 4: Pre/Post reading Reddit Posts from r/bonds Portfolio Allocations

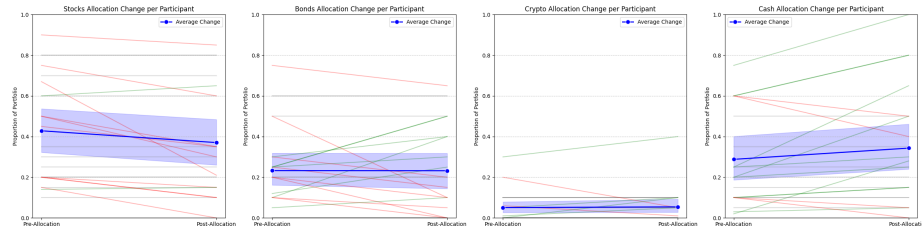


Fig. 5: Pre/Post reading Reddit Posts from r/bonds Portfolio Allocations

4.3 Crypto - Portfolio Allocations Pre/Post Reddit Influence

After assessing the two-sided paired t-test results of portfolio asset allocation differences for r/CryptoMarkets survey participants in Table 5, we see that differences in crypto and bond allocations are statistically significant at a 95% confidence level. In the boxplot chart in Figure 6, the median and upper quartile bond allocation remained the same, while the lower quartile decreased, indicat-

ing participants who had a lower allocation of bonds decreased it further. Based on the slope charts in Figure 7, we observe an average increased allocation into crypto and decreased allocation into bonds.

Table 5: Portfolio Allocation Statistics: Pre/Post reading posts from r/CryptoMarkets

Asset Class	Pre-Reddit allocation		Post-Reddit allocation		t-statistic	p-value
	Mean	Median	Mean	Median		
Stocks	0.4468	0.5000	0.4240	0.5000	1.3241	0.1979
Bonds	0.2278	0.2000	0.1920	0.2000	2.2028	0.0375
Crypto	0.0822	0.0000	0.1488	0.1000	-2.0887	0.0475
Cash	0.2432	0.1000	0.2352	0.1000	0.8109	0.4254

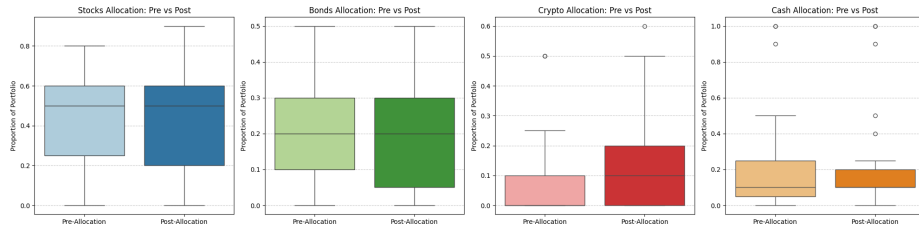


Fig. 6: Pre/Post reading Reddit Posts from r/CryptoMarkets Portfolio Allocations

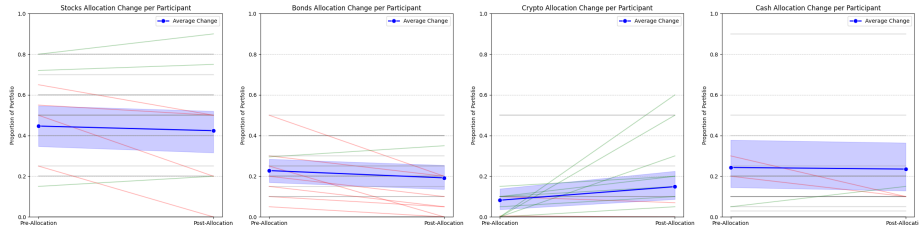


Fig. 7: Pre/Post reading Reddit Posts from r/CryptoMarkets Portfolio Allocations

4.4 Stocks - Portfolio Allocations Pre/Post LLM Influence

Anthropic Claude Sonnet 4.6, OpenAI GPT-5.3, Google Gemini 3, Microsoft Copilot, and xAI Grok were given the prompt: "What is the best asset class to invest in out of stocks, bonds, crypto, and cash? Give a one-paragraph recommendation". The results of these prompts unanimously recommended that stocks were the best asset class to invest in among the given options, and provided several reasons. Examining the portfolio allocation results before and after showing participants the prompt results revealed a statistically significant difference in allocations for stocks, crypto, and cash when conducting a two-sided paired t-test at the 95% confidence level, as illustrated in Table 6. The mean allocation in stocks rose while the allocations in crypto and cash decreased, as shown in the box plots and slope plots in Figure 8 and 9. This suggests that participants were influenced by the LLMs' recommendations, allocating a greater share of their portfolios to stocks while reducing their allocations to crypto and to cash.

Table 6: Portfolio Allocation Statistics: Pre/Post reading LLM responses

Asset Class	Pre-LLM allocation		Post-LLM allocation		t-statistic	p-value
	Mean	Median	Mean	Median		
Stocks	0.5200	0.6000	0.6400	0.7000	-3.0813	0.0051
Bonds	0.1420	0.1000	0.1720	0.1500	-1.2890	0.2097
Crypto	0.1260	0.0000	0.0740	0.0000	2.7969	0.0100
Cash	0.2120	0.2000	0.1140	0.1000	3.3372	0.0027

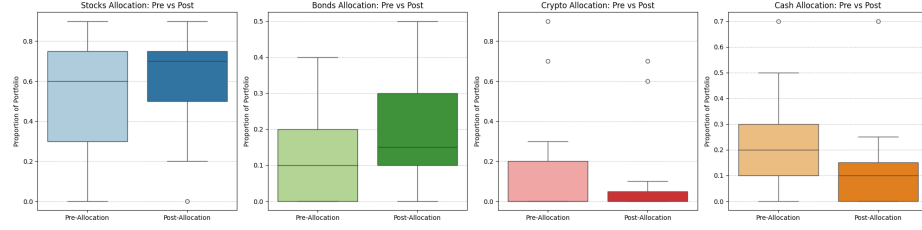


Fig. 8: Pre/Post reading LLM responses Portfolio Allocations

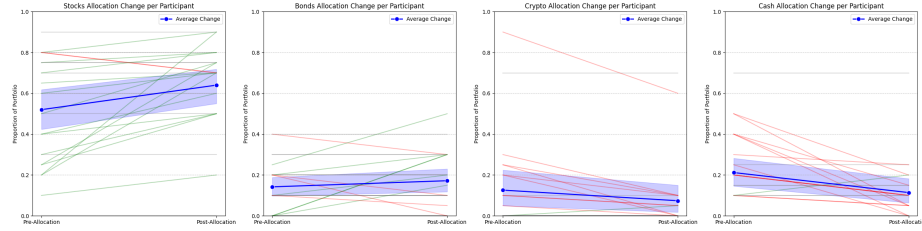


Fig. 9: Pre/Post reading LLM responses Portfolio Allocations

4.5 Portfolio Reallocation Relationships between Asset Classes

When individuals reallocate their portfolios after being presented new financial content, an understanding of how they fund their reallocation helps to understand the relationship between asset classes from a behavioral perspective. In this study, we compute the correlation of the change in asset classes using data points across surveys. Change in Stocks vs. Change in Bonds (0.13) and Change in Stocks vs. Change in Cash (0.09) are very weakly positively correlated. Change in Stocks vs. Change in Crypto (-0.41), Change in Bonds vs. Change in Crypto (-0.51), and Change in Bonds vs. Change in Cash (-0.44) are moderately correlated, suggesting that investors are likely to reallocate between those asset classes. Meanwhile, the Change in Stocks vs. Change Cash (-0.79) is strongly correlated, indicating that investors are likely to allocate between cash and stocks when rebalancing their portfolios.

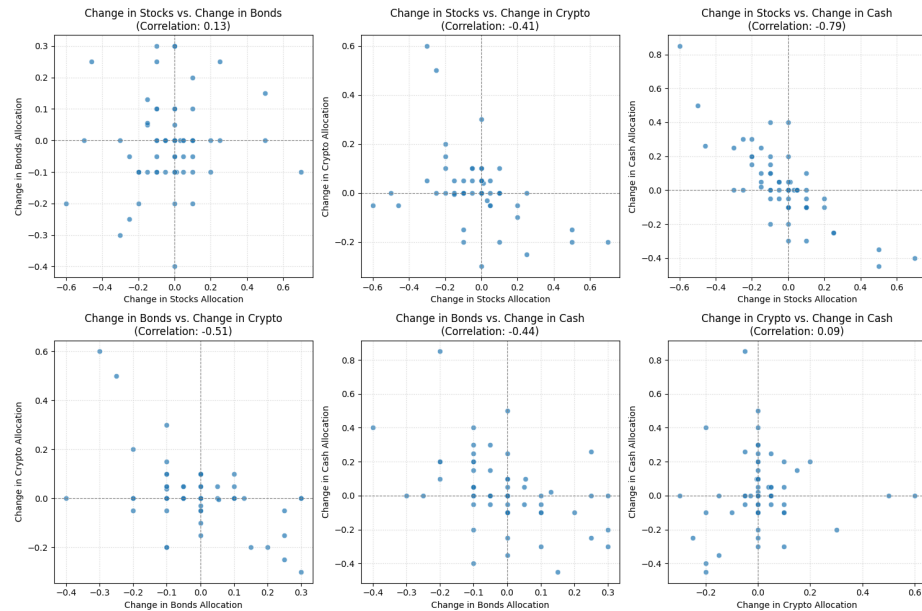


Fig. 10: Pre/Post reading LLM responses Portfolio Allocations

5 Conclusions and Recommendations

This study found that showing social media content from Reddit related to a particular asset class led to a statistically significant difference in that asset class's portfolio allocation when the asset class was either stocks or crypto. When exposed to stock market focused top posts from the past year, survey participants lowered their stock allocations and increased their cash holdings. In contrast, exposure to crypto top posts led to an increased allocation in crypto and a lowered allocation in bonds.

As with social media, content generated by LLMs can also influence individuals to change their portfolio asset allocations. This study found that exposure to prompts recommending stocks as an asset class led to an increased concentration in stocks, with decreased exposure to crypto and cash, an effect nearly opposite to that of social media exposure. These findings support the notion that social media content and LLMs can influence an individual's asset-allocation choices in their investment portfolio.

The social media aspect of this study was conducted using Reddit social media data. Future work should explore other modalities of social media content from other platforms, such as short-form videos sourced from TikTok. Furthermore, sentiment should be examined to determine how its strength and direction affect an individual's portfolio asset-allocation choices. However, sentiment can be difficult to assess because different communities may use distinct vocabularies, while sarcasm and humor are hard to detect.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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